Final Exam — Solutions

Lectures 3–4: Vector Calculus, Optimization, Probability and Statistics

Problem 1

(a) For one data point, the chain rule along $\mathbf{w} \mapsto z = \mathbf{w}^\top \mathbf{x}_i \mapsto p_i = \sigma(z) \mapsto \ell_i$ gives $\frac{d\ell_i}{dz} = (-\frac{y_i}{p_i} + \frac{1-y_i}{1-p_i}) p_i(1-p_i) = p_i - y_i$, so $\nabla \ell_i(\mathbf{w}) = (p_i - y_i)\mathbf{x}_i$. Averaging and adding $\nabla \frac{\lambda}{2}\|\mathbf{w}\|^2 = \lambda \mathbf{w}$ gives $\nabla J = \frac{1}{N}\sum_i (p_i - y_i)\mathbf{x}_i + \lambda \mathbf{w} = \frac{1}{N} X^\top (\mathbf{p} - \mathbf{y}) + \lambda \mathbf{w}$. The Jacobian of $(\sigma(\mathbf{w}^\top \mathbf{x}_i) - y_i)\mathbf{x}_i$ is $\mathbf{x}_i\, \sigma'(\mathbf{w}^\top \mathbf{x}_i)\mathbf{x}_i^\top = p_i(1-p_i)\mathbf{x}_i \mathbf{x}_i^\top$, so $\nabla^2 J = \frac{1}{N}\sum_i p_i(1-p_i)\mathbf{x}_i\mathbf{x}_i^\top + \lambda I = \frac{1}{N} X^\top D X + \lambda I$.

(b) For a unit vector $\mathbf{v}$, $\mathbf{v}^\top \nabla^2 J(\mathbf{w}) \mathbf{v} = \frac{1}{N}\sum_i p_i(1-p_i)(\mathbf{x}_i^\top \mathbf{v})^2 + \lambda$. Since $0 < p_i(1-p_i) \le \frac{1}{4}$,

$$\lambda \le \mathbf{v}^\top \nabla^2 J(\mathbf{w}) \mathbf{v} \le \frac{1}{4N}\|X\mathbf{v}\|^2 + \lambda \le \frac{\|X\|_2^2}{4N} + \lambda = L.$$

By the Rayleigh quotient, all eigenvalues of the (symmetric) Hessian lie in $[\lambda, L]$. All Hessian eigenvalues $\ge \lambda$ means that J is λ-strongly convex, and $\|\nabla^2 J\|_2 \le L$ everywhere means that J is L-smooth. A strongly convex function has exactly one minimizer.

(c) $X^\top X = 4 I_2$, so $\|X\|_2^2 = \lambda_{\max}(X^\top X) = 4$, and with $N = 4$: $L = \frac{4}{16} + 0.05 = 0.3$, $\mu = \lambda = 0.05$, $\frac{\mu}{L} = \frac{1}{6}$. By the convergence rate for strongly convex functions,

$$J(\mathbf{w}_k) - J^* \le \Big(1 - \frac{1}{6}\Big)^k (J(\mathbf{w}_0) - J^*) \le e^{-k/6}(J(\mathbf{w}_0) - J^*),$$

which is at most $10^{-6}(J(\mathbf{w}_0) - J^*)$ if $k \ge 6 \cdot 6 \log 10 \approx 82.9$. So $k = 83$ steps suffice. (With $\log \frac{6}{5}$ instead of $\frac{1}{6}$, 76 steps suffice.)

(d) Each term of J is positive, since $p_i \in (0,1)$, so $J(\mathbf{w}) > 0$ for every $\mathbf{w}$. Along $\mathbf{w} = t\mathbf{u}$ with $t \to \infty$: if $y_i = 1$, then $t\mathbf{u}^\top \mathbf{x}_i \to \infty$, so $p_i \to 1$ and $-\log p_i \to 0$; if $y_i = 0$, then $p_i \to 0$ and $-\log(1 - p_i) \to 0$. Hence $\inf J = 0$, which is not attained, and J has no minimizer. (Regularization with $\lambda > 0$ fixes this, by part (b).)

Problem 2

(a) $F_{ij} = X_{ir} B_{rj}$, so $\frac{\partial F_{ij}}{\partial X_{kl}} = \delta_{ik} B_{lj}$. By the chain rule, $(\nabla_X L)_{kl} = G_{ij} \frac{\partial F_{ij}}{\partial X_{kl}} = G_{ij}\delta_{ik} B_{lj} = G_{kj} B_{lj} = (GB^\top)_{kl}$. As a contraction: `np.einsum("ij,ijkl->kl", G, DF)`. (In practice, one computes `G @ B.T` and never forms the $mpmn$ entries of `DF`.)

(b) By Jacobi's formula, $\nabla \log\det \Sigma = \Sigma^{-\top} = \Sigma^{-1}$ ($\det \Sigma > 0$). By the formula $\nabla_X \operatorname{tr}(AX^{-1}) = -X^{-\top} A^\top X^{-\top}$ with $A = S$, $\nabla \operatorname{tr}(S\Sigma^{-1}) = -\Sigma^{-1} S \Sigma^{-1}$ (all matrices symmetric). So

$$\nabla \varphi(\Sigma) = \Sigma^{-1} - \Sigma^{-1} S \Sigma^{-1} = \Sigma^{-1}(\Sigma - S)\Sigma^{-1},$$

which is 0 if and only if $\Sigma = S$, since Σ^{-1} is invertible.

(c) $M = S^{-1/2}\Sigma S^{-1/2}$ is symmetric positive definite, with eigenvalues $m_1, \dots, m_d > 0$. Then $\log\det\Sigma = \log\det S + \sum_i \log m_i$ and $\operatorname{tr}(S\Sigma^{-1}) = \operatorname{tr}(S^{1/2}M^{-1}S^{-1/2}) = \operatorname{tr}(M^{-1}) = \sum_i \frac{1}{m_i}$, while $\varphi(S) = \log\det S + d$. Hence

$$\varphi(\Sigma) - \varphi(S) = \sum_{i=1}^{d} (t_i - 1 - \log t_i), \qquad t_i = \frac{1}{m_i}.$$

The function $h(t) = t - 1 - \log t$ is convex on $(0,\infty)$ with $h'(1) = 0$, so $h \geq h(1) = 0$, with equality only at $t = 1$. So $\varphi(\Sigma) \geq \varphi(S)$, with equality only if $M = I$, i.e., $\Sigma = S$. For the Gaussian, $-\ell(\boldsymbol{\mu}, \Sigma) = \frac{N}{2}\log\det\Sigma + \frac{1}{2}\sum_i (\mathbf{x}_i - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{x}_i - \boldsymbol{\mu}) + \frac{Nd}{2}\log(2\pi)$. At $\boldsymbol{\mu} = \bar{\mathbf{x}}$, the sum is $\operatorname{tr}\left(\Sigma^{-1}\sum_i (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^\top\right) = N\operatorname{tr}(S\Sigma^{-1})$, so $-\ell = \frac{N}{2}\varphi(\Sigma) + \text{const}$, which is minimized exactly by $\Sigma = S$.

Problem 3

(a) $g_1 = x_1 + x_2 - 1$, $g_2 = -x_2$. KKT: stationarity $2(x_1 - 3) + \mu_1 = 0$ and $2(x_2 - 1) + \mu_1 - \mu_2 = 0$; feasibility $g_1, g_2 \leq 0$; $\mu_1, \mu_2 \geq 0$; $\mu_1 g_1 = \mu_2 g_2 = 0$. Check the active sets:

- none active: $\mathbf{x} = (3, 1)$, infeasible ($g_1 = 3$);
- only g_1 active: $x_1 - 3 = x_2 - 1$ and $x_1 + x_2 = 1$ give $(\frac{3}{2}, -\frac{1}{2})$, infeasible ($x_2 < 0$);
- only g_2 active: $x_1 = 3$, $x_2 = 0$, infeasible ($g_1 = 2$);
- both active: $\mathbf{x}^* = (1, 0)$, $\mu_1 = 4$, $\mu_2 = 2(0 - 1) + 4 = 2$, both ≥ 0.

So the minimum is at $(1, 0)$, with $f = 5$, $\mu_1 = 4$, $\mu_2 = 2$.

(b) f is convex (Hessian $2I$) and g_1, g_2 are affine, hence convex. By the converse part of the KKT theorem, every KKT point is a global minimum. (Geometrically: $(1, 0)$ is the point of the feasible region closest to $(3, 1)$.)

(c) With $g(\mathbf{x}) = 1 - \mathbf{a}^\top\mathbf{x} \leq 0$, $\mathcal{L}(\mathbf{x}, \mu) = \frac{1}{2}\|\mathbf{x}\|^2 + \mu(1 - \mathbf{a}^\top\mathbf{x})$. It is convex in $\mathbf{x}$, minimized where $\mathbf{x} - \mu\mathbf{a} = \mathbf{0}$, so

$$d(\mu) = \frac{\mu^2}{2}\|\mathbf{a}\|^2 + \mu - \mu^2\|\mathbf{a}\|^2 = \mu - \frac{\mu^2}{2}\|\mathbf{a}\|^2.$$

This is maximized over $\mu \geq 0$ at $\mu^* = \frac{1}{\|\mathbf{a}\|^2}$, with $d^* = \frac{1}{2\|\mathbf{a}\|^2}$. The primal problem asks for the point of the half-space closest to $\mathbf{0}$; it lies on the hyperplane $\mathbf{a}^\top\mathbf{x} = 1$, at the distance $\frac{|0-1|}{\|\mathbf{a}\|}$ (Lecture 2), so $p^* = \frac{1}{2\|\mathbf{a}\|^2} = d^*$, attained at $\mathbf{x}^* = \mu^*\mathbf{a} = \frac{\mathbf{a}}{\|\mathbf{a}\|^2}$.

Problem 4

(a) $\operatorname{Corr}(X, Y) = \frac{3}{2\cdot 3} = \frac{1}{2}$; $\operatorname{Var}(2X - Y) = (2, -1)\Sigma(2, -1)^\top = 16 + 9 - 12 = 13$.

(b) By the conditioning formula, $Y \mid X = x \sim \mathcal{N}(\frac{3}{4}x,\ 9 - \frac{3\cdot 3}{4}) = \mathcal{N}(\frac{3}{4}x, \frac{27}{4})$.

(c) $\mathbb{E}[\operatorname{Var}(Y \mid X)] = \frac{27}{4}$ (the conditional variance does not depend on x); $\operatorname{Var}(\mathbb{E}[Y \mid X]) = \operatorname{Var}(\frac{3}{4}X) = \frac{9}{16}\cdot 4 = \frac{9}{4}$; the sum is $9 = \operatorname{Var}(Y)$.

(d) $(X, Y - \frac{3}{4}X)^\top = A(X,Y)^\top$ with the invertible $A = \begin{pmatrix} 1 & 0 \\ -\frac{3}{4} & 1 \end{pmatrix}$, so it is Gaussian with the covariance matrix $A\Sigma A^\top$, whose off-diagonal entry is $\mathrm{Cov}(X, Y - \frac{3}{4}X) = 3 - \frac{3}{4} \cdot 4 = 0$. A Gaussian vector with a diagonal (block diagonal) covariance matrix has independent components.

(e) $L = \begin{pmatrix} 2 & 0 \\ \frac{3}{2} & \frac{3\sqrt{3}}{2} \end{pmatrix}$: $L_{11}^2 = 4$, $L_{21}L_{11} = 3$, $L_{21}^2 + L_{22}^2 = 9$. For $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, I_2)$, $L\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, LL^\top) = \mathcal{N}(\mathbf{0}, \Sigma)$, i.e., $X = 2Z_1$ and $Y = \frac{3}{2}Z_1 + \frac{3\sqrt{3}}{2}Z_2$.

Problem 5

(a) $\ell(\lambda) = N \log \lambda - \lambda \sum_i x_i$, $\ell'(\lambda) = \frac{N}{\lambda} - \sum_i x_i = 0$ gives $\hat{\lambda} = \frac{N}{\sum_i x_i} = \frac{1}{\bar{x}}$, a maximum since $\ell'' = -\frac{N}{\lambda^2} < 0$.

(b) $\log p(x \mid \lambda) = \log \lambda - \lambda x$; the score is $\frac{1}{\lambda} - x$ (mean 0, since $\mathbb{E}[X] = \frac{1}{\lambda}$), and $-\frac{\partial^2}{\partial \lambda^2} \log p = \frac{1}{\lambda^2}$, so $I(\lambda) = \frac{1}{\lambda^2}$ (also $= \mathrm{Var}(X)$). Cramér–Rao: $\mathrm{Var}(\tilde{\lambda}) \geq \frac{1}{NI(\lambda)} = \frac{\lambda^2}{N}$ for unbiased $\tilde{\lambda}$.

(c) CLT: $\sqrt{N}(\bar{X} - \frac{1}{\lambda}) \to \mathcal{N}(0, \frac{1}{\lambda^2})$. With $g(t) = \frac{1}{t}$, $g'(\frac{1}{\lambda}) = -\lambda^2$, the delta method gives $\sqrt{N}(\hat{\lambda} - \lambda) \to \mathcal{N}(0, \lambda^4 \cdot \frac{1}{\lambda^2}) = \mathcal{N}(0, \lambda^2)$. The asymptotic variance $\lambda^2 = \frac{1}{I(\lambda)}$ is the Cramér–Rao bound: the MLE is asymptotically efficient. ($\hat{\lambda}$ itself is biased: $\mathbb{E}[\hat{\lambda}] = \frac{N}{N-1}\lambda$.)

(d) Posterior $\propto \lambda^{a-1}e^{-b\lambda} \cdot \lambda^N e^{-\lambda \sum_i x_i} = \lambda^{a+N-1}e^{-(b+\sum_i x_i)\lambda}$: Gamma with the shape $a+N$ and the rate $b + \sum_i x_i$. Posterior mean $\frac{a+N}{b+\sum_i x_i}$, MAP $\frac{a+N-1}{b+\sum_i x_i}$. Numbers: $\hat{\lambda} = \frac{10}{4} = 2.5$, posterior mean $\frac{12}{5} = 2.4$, MAP $\frac{11}{5} = 2.2$. The prior mean $\frac{a}{b} = 2$ is below $\hat{\lambda}$, so both Bayesian estimates are pulled from the MLE towards it; the Gamma posterior is skewed to the right, so its mode lies below its mean. As $N \to \infty$, all three agree.

Problem 6

(a) $\hat{p}$ is the mean of N i.i.d. Bernoulli(p) variables, with $\mathrm{Var}(\hat{p}) = \frac{p(1-p)}{N} \leq \frac{1}{4N}$.

(i) Chebyshev: $P(|\hat{p} - p| \geq 0.05) \leq \frac{1}{4N \cdot 0.0025} = \frac{100}{N} \leq 0.05$ if $N \geq 2000$.

(ii) Hoeffding (values in $[0,1]$): $2e^{-2N \cdot 0.0025} \leq 0.05 \iff N \geq \frac{\log 40}{0.005} \approx 737.8$, so $N = 738$ suffices.

(iii) $\hat{p} \approx \mathcal{N}(p, \frac{p(1-p)}{N})$; we need $1.96\sqrt{p(1-p)/N} \leq 0.05$, worst at $p = \frac{1}{2}$: $N \geq \frac{3.84 \cdot 0.25}{0.0025} \approx 384$ (385 with $1.96^2 = 3.8416$). The CLT is a statement about the limit $N \to \infty$ and gives no bound on the approximation error for a fixed N (which is large for p near 0 or 1), so this N is not guaranteed to work; (i) and (ii) are guarantees.

(b) $\mathrm{KL}(p \,\|\, q) = \log \frac{2}{1} + \frac{1+(0-1)^2}{2 \cdot 4} - \frac{1}{2} \approx 0.693 + 0.25 - 0.5 = 0.443$. $\mathrm{KL}(q \,\|\, p) = \log \frac{1}{2} + \frac{4+1}{2} - \frac{1}{2} \approx -0.693 + 2 = 1.307$. The second is larger: $\mathrm{KL}(q \,\|\, p) = \mathbb{E}_q[\log \frac{q}{p}]$ averages over samples of the wide q, which often land in the tails of the narrow p, where $\log \frac{q}{p}$ is large (it grows like $\frac{x^2}{2}$). In $\mathrm{KL}(p \,\|\, q)$, the wide q covers all the mass of p, and the ratio stays moderate. KL divergence is not symmetric.